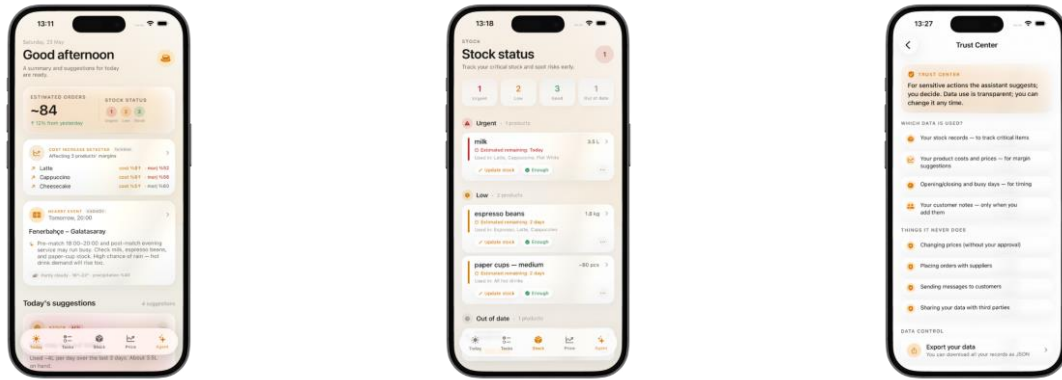


AI Integration in Turkish Microbusinesses

Challenges, Solutions, and a Pilot-Ready Playbook

Whitepaper

Koç University CEMS MSc in International Management · August 2026



Project Team

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Executive Summary

This whitepaper converts the project’s 41 in-person interviews and final presentation into a more rigorous, pilot-ready operating model. The evidence points to a clear opportunity: owners are not asking for abstract artificial intelligence. They are asking for fewer stock surprises, better staffing decisions, faster pricing updates, simpler accounting work, and more reliable customer follow-up.

41 owner interviews	7 business domains	4 Istanbul districts
26/41 would pay for concrete value	18/41 use AI in business today	6/41 run an integrated software stack

The strategic conclusion

Start with digitally ready cafés and restaurants, integrate with the owner’s existing POS rather than replacing it, lead with time and money rather than “AI,” and validate the proposition through a controlled pilot with 10-15 reference owners.

The source materials require a clear distinction between respondent counts and multi-label value-reason mentions. The raw field workbook contains 26 direct “yes” answers and 15 “no” answers to willingness to pay, while the presentation’s figure of 30 positive responses represents multi-label value reasons rather than 30 unique owners. The commercial conclusion remains strong: 63% of interviewed owners expressed willingness to pay when the benefit is concrete.

Field evidence translated into a mobile-first operating assistant for microbusiness owners.

The whitepaper moves from the research question and evidence base to sample findings, beachhead segmentation, product design, trust, go-to-market, pilot measurement, and supporting evidence tables.

1. Opportunity and Research Question

The project began from a diffusion gap. As reported in the final presentation's desk research, microbusinesses account for 94.2% of Turkish enterprises and productivity per employee is substantially lower than in medium-sized firms. The presentation also cites an AI-adoption rate of approximately 6.6% for enterprises with 10–49 employees. Because that statistic excludes businesses with fewer than 10 employees, it is used here only as directional context and not as a measure of AI adoption among the microbusinesses studied.

Source note: market-context figures are reproduced from the supplied final presentation and have not been independently verified in this revision. They should be treated as directional context until the original external sources, definitions, dates, and population coverage are confirmed before publication.

The project therefore moved from market context to a field question: How can Microsoft design a scalable strategy to accelerate the adoption of AI solutions among microbusinesses in Türkiye based on validated client needs? The team narrowed the scope through three choices: service-oriented rather than asset-heavy production, business-to-consumer rather than business-to-business, and perishable rather than durable output. This led to cafés, restaurants, food retailers, and street-food businesses as the primary field.

Why this scope matters

Perishable-service businesses combine frequent transactions, time-sensitive inventory, staff-intensive service, and recurring customer demand. Their operating pain is visible every day, while their digital systems remain shallow enough for a focused agent to create a noticeable difference.

2. Research Design and Data Integrity

The field study used a common six-question script in 41 structured, in-person interviews. The questions covered daily challenges, repetitive work, current digital tools, willingness to pay, AI awareness and use, and concerns about sharing business data. Responses were coded on a multi-label basis, which means one owner could contribute to several categories within the same question.

Design element	Field-study specification
Population in scope	Owner-led microbusinesses with fewer than 10 employees in B2C perishable-service domains.
Geography	Kadıköy, Şişli, Beşiktaş, and Sarıyer, covering both sides of Istanbul.
Domains	Cafés, restaurants, street vendors, greengrocers, patisseries, pubs, and supermarkets.
Protocol	Six consistent questions asked in person.
Coding	Multi-label thematic coding for Q1-Q6; owner-level cross-tabs derived from direct responses.
Interpretation	Descriptive and exploratory. The sample is purposive, not statistically representative of all Turkish microbusinesses.

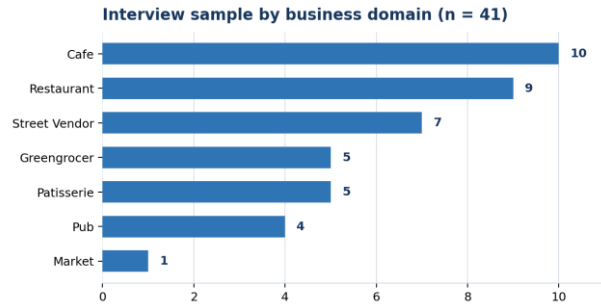
Sources: Project field-study materials, June 2026.

Respondents versus mentions

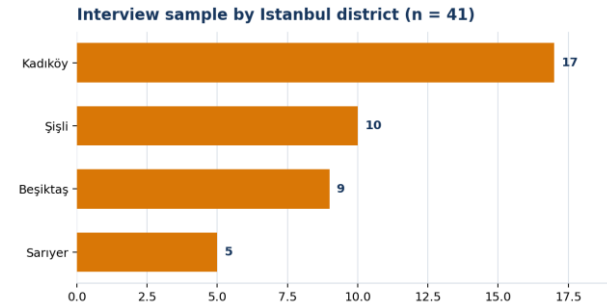
Owner-level statements answer “how many businesses said yes or use AI?” Multi-label mention counts answer “which reasons or pain points appeared, and how often?” Keeping these units separate prevents category totals from being mistaken for respondent totals.

3. Sample Profile

The sample is concentrated in cafés, restaurants, and street vendors, which together account for 26 of 41 interviews (63%). Kadıköy has the highest interview density, followed by Şişli and Beşiktaş. The age profile is relatively balanced across 25-34 and 35-44, with smaller groups in the two older ranges.



Source: Field Research-EN.xlsx. Counts are businesses.



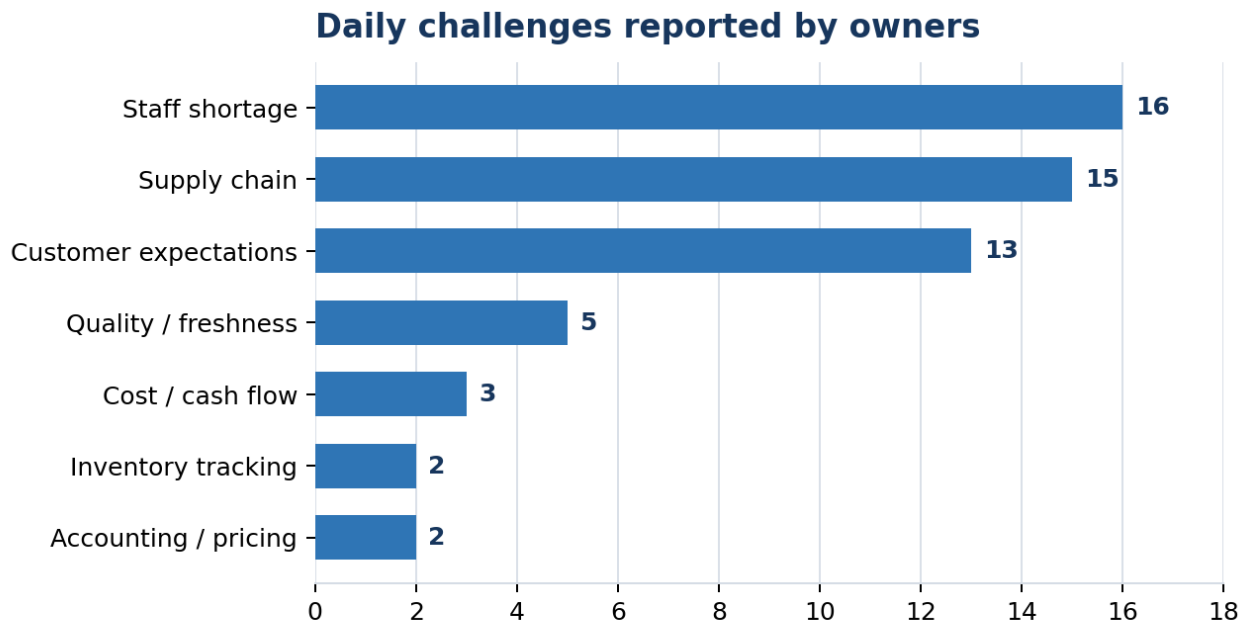
Source: Field Research-EN.xlsx. Counts are businesses.

Age range	Interviews	Would pay	Uses AI in business
25-34	13	10/13 (77%)	5/13 (38%)
35-44	14	8/14 (57%)	9/14 (64%)
45-54	8	5/8 (62%)	3/8 (38%)
55-64	6	3/6 (50%)	1/6 (17%)

Source: Project field-study data. Percentages are within the interview sample.

4. What the Field Said

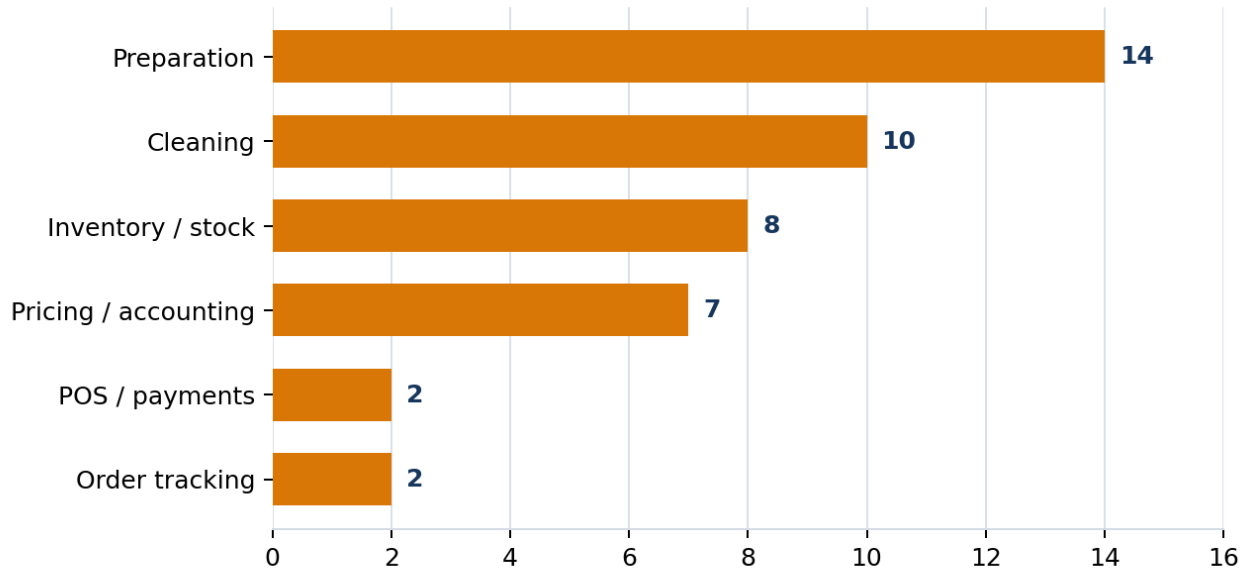
The strongest pattern is that owners describe execution problems, not strategic planning problems. Staffing, supplier reliability, and customer expectations dominate daily pressure. The tasks consuming time are similarly practical: preparation, cleaning, stock control, and pricing or accounting.



Source: Final presentation appendix, Q1. Multi-label mentions; one owner may appear in more than one category.

Figure 1. Daily operating pain is concentrated in workforce, supply, and customer pressure.

Most repetitive or time-consuming tasks



Source: Final presentation appendix, Q2. Multi-label mentions.

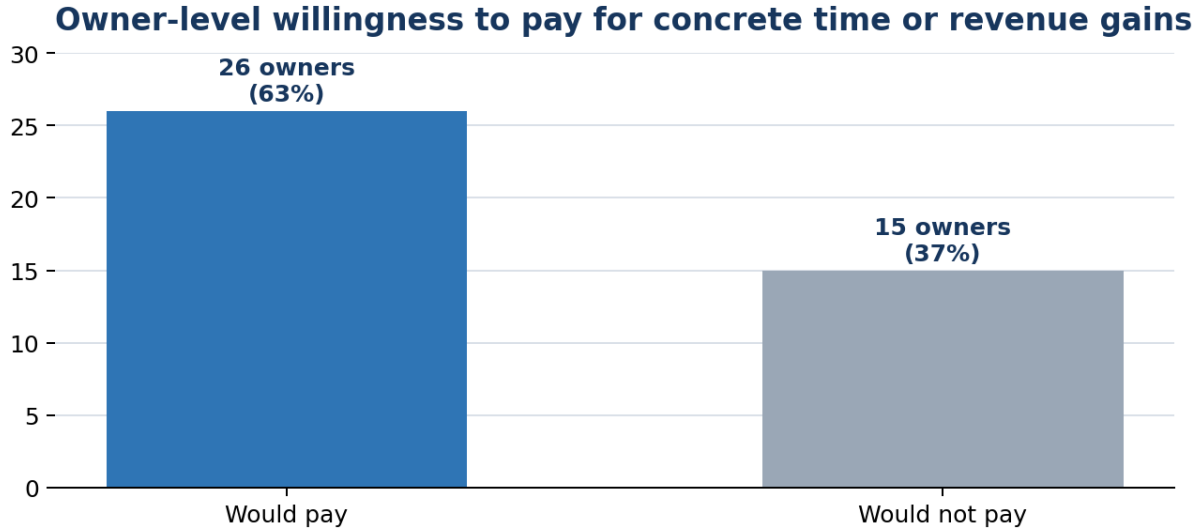
Figure 2. Repetitive work sits mainly behind the counter, not in customer-facing interaction.

Field voice	Product implication
"Stock control and product arrangement take a lot of time." — Greengrocer, Kadıköy, 55-64	Make inventory visibility immediate and low-effort; emphasize reorder timing and waste.
"Accounting, expense categorization, or tax-related processes" would justify payment. — Street vendor, Kadıköy, 35-44	Back-office assistance can be an entry use case for owners who do not identify with "AI."
"Peak hour" staffing problems live in the owner's head. — Café and restaurant pattern	Use POS demand signals to surface staffing risk before the rush.

Excerpts are anonymized and lightly shortened from the field interviews.

5. Digital Readiness, Willingness, and AI Use

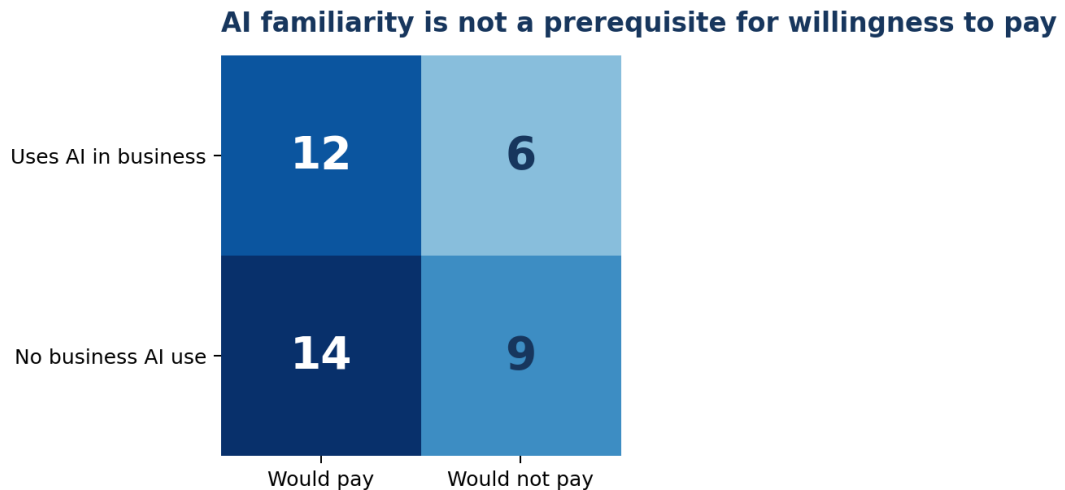
Digital adoption is broad but shallow. The presentation reports that only three businesses use no digital tools, while the dominant pattern is a POS system plus a single add-on such as a QR menu, delivery platform, or order-management tool. Only six businesses operate a more integrated stack. Within this sample, these patterns support testing the POS as a practical integration point rather than assuming that owners should replace their existing systems.



Source: Field Research-EN.xlsx, direct Q4 responses. Positive value reasons generated 30 multi-label mentions in the presentation.

Figure 3. Nearly two-thirds of owners directly expressed willingness to pay.

The raw workbook shows 26 direct “yes” responses and 15 “no” responses. Within the positive group, owners repeatedly tied value to time savings, revenue or profitability, and accounting or cost control. Within the negative group, “no perceived need” is more common than explicit distrust or price resistance. In this sample, perceived relevance and proof of concrete value appear to be important adoption factors alongside awareness.



Source: Field Research-EN.xlsx. Business-use coding counts only respondents who described using AI for business tasks.

Figure 4. Fourteen non-users of AI still said they would pay for a valuable tool.

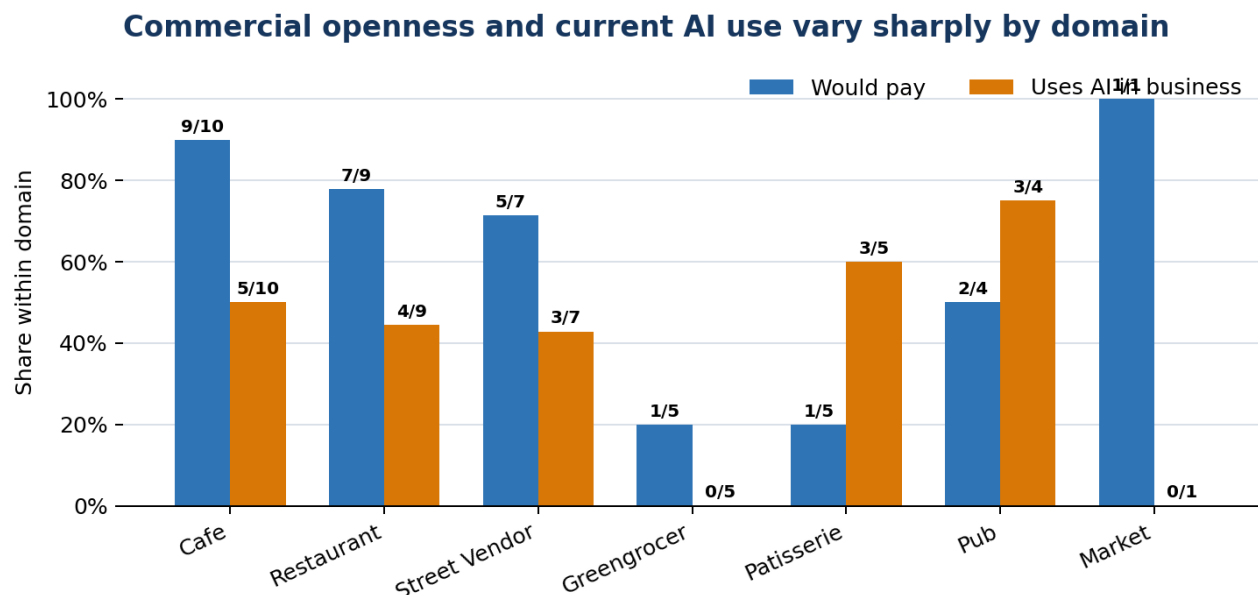
Eighteen owners described using AI for business tasks, but usage is concentrated in menu visuals, social posts, promotional copy, design support, and occasional brainstorming. Operational usage in stock, scheduling, pricing, accounting, and supply management was rarely reported in these interviews. Within this sample, the findings identify a potential area for product testing: owners already recognize AI as a creative assistant but reported little use of it as an operating layer.

Adoption insight

Current AI use is not required for willingness to pay: 14 owners who did not report business use of AI still said they would pay for a valuable tool. This cross-tab supports defining the beachhead by operational need, digital readiness, and openness to measurable value rather than by AI familiarity alone; the pilot should test which of these factors best predicts adoption and retention.

6. Beachhead Segmentation

The final presentation identifies cafés and restaurants as the proposed starting point, and the raw workbook provides supporting evidence within the sample. Nine of ten café owners and seven of nine restaurant owners expressed willingness to pay. By contrast, only one of five greengrocers and one of five patisseries did so. These small subgroup results support testing a business-type-led beachhead in the pilot, but they do not establish population-level differences.



Source: Field Research-EN.xlsx. Small subgroup sizes; percentages are descriptive, not population estimates.

Figure 5. Cafés combine the strongest willingness to pay with meaningful current AI familiarity.

Segment	Evidence in sample	Recommended approach	Role in rollout
Launch: cafés	9/10 willing to pay; 5/10 use AI in business; strongest fit with staffing, stock, pricing, and customer use cases.	POS-integrated pilot, daily briefing, stock and shift alerts.	Primary reference-owner pool.
Launch: restaurants	7/9 willing to pay; 4/9 use AI; high staff and supply pressure.	Pilot alongside cafés with staffing, supply, and margin features.	Second primary pool.
Selective: street vendors	5/7 willing to pay; accounting and customer expectations are recurring themes.	Offer simpler back-office and profitability workflows.	Targeted test cell.
Educate: pubs and patisseries	Higher creative AI usage, but weaker willingness to add another paid tool.	Prove a specific recurring problem before upsell.	Case-study-led expansion.
Later: greengrocers / supermarket	Low current AI use and low willingness, despite clear stock and supply pain.	Assisted onboarding, low-friction Base tier, concrete waste and reorder proof.	Later adoption segment.

Derived from the project's field-study data. Small subgroup sizes require pilot validation.

Geographically, the presentation recommends Kadıköy and Şişli because they combine interview density and stronger digital stacks. The raw responses also show high willingness in Beşiktaş (8 of 9), making it a useful second-wave comparison area after the initial district pair.

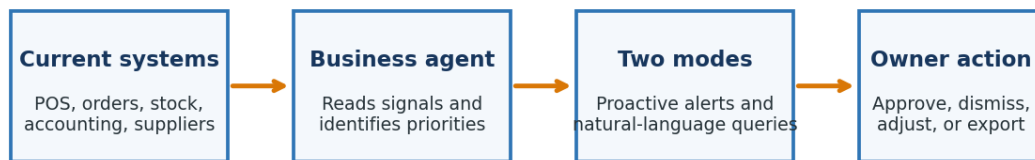
7. Product Blueprint: Five Operational Problem Areas

The proposed product is a Microsoft-branded, mobile-first business agent that sits above the owner's existing systems. It should not ask the owner to become an analyst. It should convert business signals into a short list of decisions that can be understood and acted on during a busy day.

Problem area	Field evidence	Agent behavior	Suggested sequence
Staffing and scheduling	Staff shortage: 16 mentions, the top daily challenge.	Detect peak hours from POS transactions, generate shifts, warn before understaffing.	MVP
Supply-chain alerts	Supply disruption: 15 mentions.	Compare delivery patterns with stock risk; surface alternatives before a stockout.	Phase 2
Inventory and stock	Inventory/stock: 8 repetitive-task mentions.	Track thresholds, flag low items, recommend reorder timing, highlight waste.	MVP
Dynamic pricing / accounting	Pricing/accounting: 7 repetitive-task mentions.	Flag margin erosion, suggest price updates from cost and demand signals, support owner-controlled frequency.	MVP
Customer tracking	Customer expectations: 13 mentions; operational AI use near zero.	Recognize loyal customers, record preferences, prepare owner-approved outreach.	Phase 2

Evidence counts and product behaviors are drawn from the final presentation. MVP sequencing is a proposed prioritization based on immediacy and POS-data fit.

Mobile-first AI agent operating on top of the current workflow

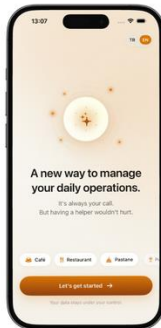


Trust is built into the loop: visible data use, owner approval for sensitive actions, and no automatic third-party sharing.

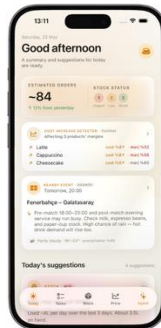
Figure 6. The agent adds an intelligence layer without replacing the owner's current workflow.

8. Agent Experience and Trust Model

The prototype translates the strategy into a mobile experience. Onboarding learns the business type, priorities, key products, and operating rhythm. The home screen then consolidates risks, opportunities, and actions. Dedicated views cover tasks, stock, customers, pricing, integrations, performance, and trust controls.



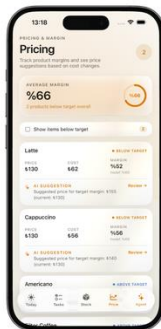
Onboarding



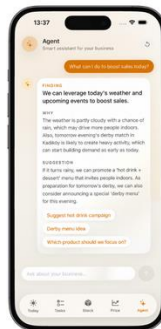
Daily briefing



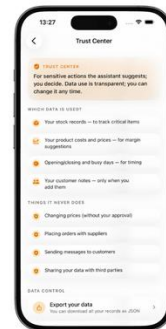
Stock status



Pricing



Agent reasoning



Trust Center

Figure 7. Selected screens from the mobile-first Swift prototype in the final presentation.

Source: Project presentation, slides 22-31.

Proactive first, interactive when needed

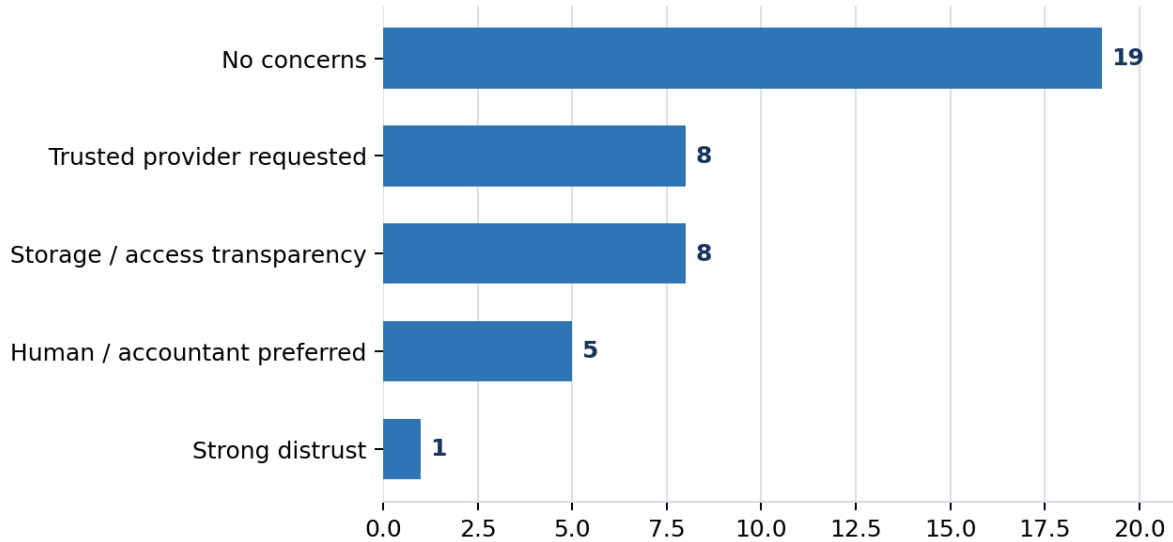
Owners repeatedly described days dominated by preparation, cleaning, service, and stock checks. The agent should therefore lead with short proactive alerts instead of requiring repeated queries. Example alerts from the presentation include “Stock of item X below threshold - reorder?”, “Peak hour in two hours - you have two staff scheduled,” and “Supplier X delivery is two days late.”

Interactive mode adds natural-language access to the business: “How is my tomato stock?”, “What is my expected customer count today?”, or “What is my most-sold item this month?” The agent may ask at most one single-tap calibration question per day so that learning does not become another task.

Trust as a product feature

The field does not show a blanket refusal to share data. Nineteen coded mentions reported no concern, while most remaining concerns request a trusted provider, transparent storage and access, or a responsible human contact. Only one response was coded as strong distrust. The product should answer these requests visibly rather than burying them in policy text.

What owners said about business-data sharing



Source: Final presentation appendix, Q6. Multi-label mentions.

Figure 8. Most trust concerns are concrete design requirements, not an adoption wall.

Trust requirement	Experience response
Recognizable accountability	Microsoft branding at the product surface and a clear support path.
Transparent data use	Show which records are used and why: stock, costs, operating hours, and owner-added customer notes.
Owner approval	Never change prices, place supplier orders, or send customer messages without approval.
Data control	Allow permissions to be changed, records to be exported, and connections to be removed.
Human in the loop	Keep accountants or responsible staff compatible with the workflow rather than positioning the agent as a replacement.

9. Go-to-Market and Packaging

The go-to-market principle is to meet owners where they already spend attention and let the product be discovered through familiar channels. The final presentation proposes Instagram and TikTok for discovery, Google Search and Maps for intent, and WhatsApp peer groups for trust and word of mouth. Positioning should lead with hours saved, revenue lifted, or accounting hassle removed - never with AI as an abstraction.

Channel	Role	Example execution	Success signal
Instagram / TikTok	Discovery	Short Turkish videos showing one daily problem solved; district and business-category targeting.	Qualified installs and demo starts.
Google Search / Maps	Intent capture	Operational keywords around stock, pricing, staffing, and POS tools; App Store optimization.	High-intent trial activation.
WhatsApp peer networks	Trust and retention	Reference owners share named case-study cards and earn referral benefits.	Referral conversion and retention.
Microsoft product surface	Accountability	Visible branding, support, permissions, and data controls on every critical screen.	Data-connection completion and trust scores.

Three tiers of value

Tier	Position	What it delivers	Commercial goal
Base	Free or near-free trust builder	Daily briefing, generic stock and shift heuristics, natural-language questions; no POS integration required.	Convert the “no perceived need” segment.
Core	POS-connected operating engine	Live stock and sales signals, reorder timing, shift planning, and customer insights.	Create the “this saves me hours” moment.
Premium	Multi-system intelligence	Supplier performance, alternate sourcing, dynamic pricing, loyalty campaigns, and demand-based scheduling.	Become the operating layer of the business.

10. Pilot Roadmap and Measurement

The recommendation is a gated 24-month sequence rather than a broad national launch. It begins with a controlled pilot, expands only after evidence of accuracy, trust, retention, and commercial value, and proceeds to national replication only if those gates are met.

Phase	Timing and focus	Key moves	Gate to advance
1. Digital seed	Months 1-4 · controlled pilot	Onboard 10-15 reference owners with free Premium; acquire 50 active Base users; validate alerts; produce 3-5 named case studies.	Owner recommendation score at least 7/10 and at least 3 validated case studies.
2. Validate and scale	Months 5-10 · public expansion	Activate WhatsApp referrals; launch POS integrations; market Core after the Base trial; expand to patisseries, street vendors, and greengrocers.	POS-connected users retain at least 2× better than Base-only users.
3. National coverage	Months 11-24 · optimize and deepen	Use Istanbul cases nationally; pursue App Store and SME-media visibility; scale Premium and document the playbook.	Validated acquisition playbook with proven unit economics, ready to replicate across Türkiye.

Pilot measurement framework

Question	Metric	Why it matters
Are alerts reliable?	Alert accuracy and owner-confirmed false positives.	Inaccurate proactive advice destroys trust quickly.
Do owners act?	Alert open rate, action rate, dismiss rate, and time to action.	Measures whether the agent fits daily workflow.
Is value visible?	Owner-reported time saved; avoided stockouts; margin or sales proxy where available.	Connects the product to the concrete value language found in Q4.
Is trust sufficient?	Integration completion, permission changes, data-control usage, and trust feedback.	Tests whether transparency converts caution into adoption.
Is the product commercially viable?	Base-to-Core conversion, retention by integration depth, referral conversion.	Validates packaging and the role of POS connectivity.
Can the story travel?	Number of named, measurable case studies and referral-led acquisitions.	Builds the repeatable word-of-mouth playbook.

Immediate next step

Develop the pilot-ready agent, recruit 10-15 reference owners in Kadıköy and Şişli, provide free Premium access during the test, and use the first four months to produce accurate alerts, measurable owner outcomes, and at least three credible case studies.

11. Limitations and Conclusion

Limitations

- The sample is purposive and concentrated in four relatively central Istanbul districts; it is not representative of all Turkish microbusinesses.
- The research focuses on B2C perishable-service firms, so findings should not be generalized to asset-heavy production or B2B sectors without further fieldwork.
- Willingness to pay is self-reported intention, not observed conversion or retention.
- Some presentation metrics are multi-label mention counts. The whitepaper separates those counts from unique-owner counts wherever the raw workbook permits.
- The prototype demonstrates product logic and experience, but no live integration, pilot accuracy, or unit-economics results are included in the supplied materials.
- Subgroup cross-tabs have small cell sizes and should be used for pilot design, not population inference.

Conclusion

The field evidence supports a focused opportunity rather than a broad AI campaign. Microbusiness owners are already surrounded by digital tools, but their systems do not relieve the operational burdens that dominate their days. A mobile agent connected to existing POS data can close that gap by surfacing stock risk, staffing needs, margin pressure, supplier delays, and customer opportunities before they become problems.

The strongest commercial opening is among cafés and restaurants, where willingness to pay is highest and current digital behavior is sufficiently mature. The proposition should be framed in the owner's language: hours saved, fewer mistakes, better margins, and less mental load. Trust should be made visible through brand accountability, transparent data use, and owner approval for consequential actions.

The project therefore reduces to one testable decision: build a narrow, proactive, trust-centered operating agent and validate it with 10–15 reference owners. A successful pilot could show whether the field evidence can be translated into a repeatable playbook for broader adoption across Türkiye.

Appendix: Evidence Tables

A1. Question-by-question synthesis

Question	Leading findings	Interpretation
Q1 · Daily challenges	Staff shortage 16; supply chain 15; customer expectations 13.	The dominant problems are operational execution issues.
Q2 · Repetitive tasks	Preparation 14; cleaning 10; inventory 8; pricing/accounting 7.	Back-of-house workload is the clearest automation territory.
Q3 · Digital tools	POS plus one add-on dominates; 6 integrated stacks; 3 use no tools.	Integrate with the POS rather than replacing the current system.
Q4 · Willingness to pay	26 of 41 direct yes responses; 30 positive value-reason mentions; 15 direct no responses.	Concrete time, revenue, and accounting value is the unlock.
Q5 · AI use	18 use AI in business; usage is mainly creative and promotional.	Operational AI remains largely unoccupied.
Q6 · Data sharing	19 no-concern mentions; concerns focus on provider trust, transparency, and human accountability.	Trust is a product requirement, not a universal blocker.

A2. Segment cross-tab

Domain	n	Would pay	Uses AI in business
Café	10	9/10 (90%)	5/10 (50%)
Restaurant	9	7/9 (78%)	4/9 (44%)
Street Vendor	7	5/7 (71%)	3/7 (43%)
Greengrocer	5	1/5 (20%)	0/5 (0%)
Patisserie	5	1/5 (20%)	3/5 (60%)
Pub	4	2/4 (50%)	3/4 (75%)
Supermarket	1	1/1 (100%)	0/1 (0%)

Source: Project field-study data. Business-use AI coding excludes respondents who described personal-only use.

Acknowledgements

This whitepaper synthesizes the field research and final recommendations delivered by the Koç University student team in June 2026 under the CEMS MSc in International Management program. The project was coached by Barbaros Günay of Microsoft Türkiye and supervised by Senior Lecturer Dr. Melike Mermercioğlu of Koç University. The student team comprised Naz Genç, Caner Ulaş Subaşı, and Emre Can Akdömbek of Koç University, with Driss Schmid of WU Vienna.

Field-study figures in this revision are drawn from the project's field-study materials and presentation. Market-context figures are identified separately and require verification against their original sources before publication.